Open-Source Resources for Sepsis Detection and Alerting Systems

This document provides a curated list of open-source projects, AI models, and technical implementations related to sepsis detection and early warning systems, which can be valuable for developing the SEWA application.

1. Machine Learning Models for Sepsis Prediction

- **redhat-na-ssa/sepsis-detection** [1]
    - **Description:** An online containerized machine learning microservice designed to predict patients likely to become septic based on vital signs and laboratory values. The repository includes a detailed machine learning checklist covering problem framing, data acquisition (from Kaggle), exploration, preparation, model selection (SVM, Naive Bayes, Random Forest, Logistic Regression, SGD, Neural Network, XGBoost), fine-tuning, and deployment considerations. It uses Python and Jupyter Notebooks.
    - **Link:** https://github.com/redhat-na-ssa/sepsis-detection
- **nevermind78/Early-Prediction-of-Sepsis-From-Clinical-Data** [2]
    - **Description:** This project focuses on developing a machine learning-based model for the early and real-time prediction of sepsis from clinical data in intensive care units. It utilizes data from the PhysioNet/Computing in Cardiology Challenge 2019, covering various clinical time series data (vital signs, laboratory values, demographics). The repository includes Jupyter Notebooks for Exploratory Data Analysis (EDA), MLP models, and a presentation.
    - **Link:** https://github.com/nevermind78/Early-Prediction-of-Sepsis-From-Clinical-Data
- **ikoghoemmanuell/Sepsis-Prediction-with-ML-and-FastAPI** [3]
    - **Description:** A comprehensive machine learning project for sepsis prediction using classification techniques, with a focus on deploying the model as a user-friendly API using FastAPI. It highlights the importance of early recognition and intervention in sepsis management. The repository provides setup instructions for running the FastAPI application locally.
    - **Link:** https://github.com/ikoghoemmanuell/Sepsis-Prediction-with-ML-and-FastAPI

2. ICU Monitoring and Alerting Systems

- **cmudig/sepsis-reasoning-dashboard** 4
  - **Description:** This repository demonstrates a Sepsis Reasoning Cue Dashboard, showcasing model training, patient selection, and dashboard design. It is associated with a research paper on intelligent reasoning cues in complex decisions. The setup involves a Flask server and a client-side application, with model training details in Jupyter Notebooks.
  - **Link:** https://github.com/cmudig/sepsis-reasoning-dashboard
- **Sumeet2005/smart-icu-dashboard** 5
  - **Description:** An AI-powered ICU patient monitoring dashboard offering real-time sepsis risk prediction, vital signs tracking, and clinical decision support. It is built with Python, Streamlit, XGBoost, Pandas, NumPy, and Plotly, providing a practical example of a real-time monitoring interface.
  - **Link:** https://github.com/Sumeet2005/smart-icu-dashboard
- **iozwang/homeicu** 6
  - **Description:** HomeICU is an open-source remote patient monitor that uses wearable sensors to measure vital signs, enabling remote diagnosis and treatment. While not exclusively for sepsis, its architecture for real-time patient monitoring, including hardware (ESP32 SoC module) and software (Flutter/Dart for mobile app, MQTT for data transmission), is highly relevant for building an ICU monitoring system.
  - **Link:** https://github.com/iozwang/homeicu

3. Relevant Research and Implementations

- **Sepsis Watch™: the implementation of a Duke-specific early warning system for sepsis** 7
  - **Description:** This project from Duke Health describes the development and implementation of a deep learning model into clinical care to provide an early warning system for patients at risk of sepsis. It includes EHR data cleaning, a pre-launch model optimization tool, a dashboard user interface, and a quality improvement tool. This provides a real-world example of a deployed system.
  - **Link:** https://dihi.org/project/sepsiswatch/

References

[1] redhat-na-ssa. (n.d.). sepsis-detection. GitHub. Retrieved from

[2] nevermind78. (n.d.). Early-Prediction-of-Sepsis-From-Clinical-Data. GitHub. Retrieved from

[3] ikoghoemmanuell. (n.d.). Sepsis-Prediction-with-ML-and-FastAPI. GitHub. Retrieved from

[4] cmudig. (n.d.). sepsis-reasoning-dashboard. GitHub. Retrieved from

[5] Sumeet2005. (n.d.). smart-icu-dashboard. GitHub. Retrieved from

[6] iozwang. (n.d.). homeicu. GitHub. Retrieved from

[7] Duke Institute for Health Innovation. (n.d.). Sepsis Watch™: the implementation of a Duke-specific early warning system for sepsis. Retrieved from